Lasso regression :-

① What is lasso regression ?

→ Lasso regression is a type of linear regression with regularization.

→ It adds a penalty to the model coefficients to prevent overfitting.

→ Lasso stands for :- Least Absolute shrinkage & Selection operator.

→ Main purpose of this algorithm :-

↳ Reduce overfitting

↳ performs automatic feature selection.

↳ Make the model simpler.

② Problem with linear regression :-

→ Linear regression tries to minimize the error :-

$$\left[RSS = \sum (y - \hat{y})^2 \right]$$

Where → y = actual value

$\hat{y}$ = Predicted value.

the problem is model may overfit the training data.
→ All features are used even if some are not useful.

③ Idea of Regularization :-

Regularization adds a penalty term to the loss function.

- → Reduce large coefficients.
- → Prevent overfitting.
- → Improve generalization.

④ Lasso Regression Loss function :-

~~lo~~ Lasso adds L_1 penalty to linear regression.

$$\left[\text{Loss} = RSS + \lambda \sum |\beta|\right]$$

where, RSS ⇒ Residual Sum of Squares.

β ⇒ Model coefficients.

λ ⇒ regularization strength.

⑤ Meaning of Lambda (λ)

Lambda controls how strong the penalty is

$\lambda = 0$ → same as linear regression.

Small λ → Slight regularization

Large λ → Strong regularization

⑥ When to use Lasso :-

↳ Dataset has many features.
↳ Some features are irrelevant.
↳ If we want automatic feature selection.

For ex :- Marketing data, Genomics data
Text features.

⑦ Important requirement :- Feature Scaling required.

Before applying Lasso.

Because Lasso penalizes coefficients based on magnitude.

→ use :-
↳ standard Scaler.

Ridge	Lasso.
L_2 regularization	L_1 regularization.
Penalty = β^2	Penalty = $\lvert\beta\rvert$
shrink coeff	Some coeff becomes - 0
No feature selection	Feature selection.